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Direct synthesis of indazole derivatives *via* Rh(III)-catalyzed C–H activation of phthalazinones and allenes†

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A novel Rh(III)-catalyzed annulation of phthalazinones or pyridazinones with various allenes was developed, leading to the formation of indazole derivatives bearing a quaternary carbon in moderate to good yields. The targeted products were synthesized *via* sequential C–H activation and olefin insertion, followed by β-hydride elimination and intramolecular cyclization. The synthetic protocol proceeded efficiently with broad functional group tolerance, high atom efficiency and high *Z*-selectivity. The practicability of this method was proved by synthetic transformation.

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Introduction

Nitrogen-containing heterocycles represent one of the most prominent heterocyclic scaffolds with their widespread distribution in natural products and their close connection with drugs.[1] Among them, indazoles and their derivatives have privileged structures for potential biological activity.[2] As shown in Fig. 1, nigellidine has the capacity to regulate glucose consumption,[3] a PDE_4/TNFR inhibitor shows anti-HAT activity,[4] a human immunodeficiency virus protease (HIVPR) inhibitor displays antiviral activity,[5] a tyrosine threonine kinase (TTK) inhibitor possesses good anti-cancer activity,[6] and several indazolones are reported to show anti-inflammatory activities.[7] Therefore, it is of great significance to develop a simple and efficient method to synthesize indazole scaffolds.

Over the past decades, transition-metal-catalyzed C–H bond activation has been considered as an increasingly significant tool in organic synthesis, which can construct complicated molecules through synchronous C–C and C–N bond formation.[8] In particular, selective functionalization of $C(sp^2)$–H bonds assisted by the CONHR group is an appealing area.[9] For instance, transition-metal-catalyzed cyclization of *N*-aryl phthalazinones or pyridazinones with different coupling agents continues to attract wide interest.[10] In 2018, Ji and co-workers reported a Rh(III)-catalyzed oxidant-free [4 + 1] annulation reaction of *N*-aryl phthalazinones/pyridazinones and propargyl alcohols for the formation of divergent heterocycles (Scheme 1, eqn (1)).[11] Afterward, Sakhuja and co-workers developed a tandem C–H/N–H activation reaction between *N*-aryl phthalazinones and α-diazo carbonyl compounds to afford two different products by changing the types of additives (eqn (2)).[12] Moreover, they also described Ru(II)-catalyzed carbocyclization of *N*-aryl phthalazinones with aryl isocyanates (eqn (3)).[13] Based on the above discussion and our previous research in the field of transition metal-catalyzed C–H bond activation,[14] herein, we disclosed an efficient Rh(III)-catalyzed oxidative cyclization of *N*-aryl phthalazinones and pyridazinones with various allenes to afford indazole derivatives.

Results and discussion

Initially, 2-phenyl-2,3-dihydrophthalazine-1,4-dione **1a** and buta-2,3-dien-1-ylbenzene **2a** were chosen as the model substrates to achieve the optimized reaction conditions (Table 1). The substrates were treated with $[Cp^*RhCl_2]_2$ (2.5 mol%) and AgOAc (1.0 equiv.) in MeCN (1.0 mL) at 100 °C for 12 h. The

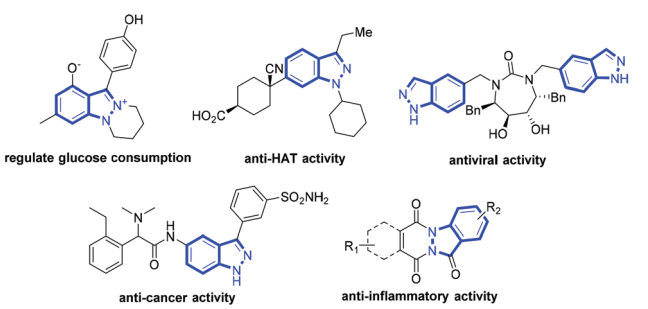
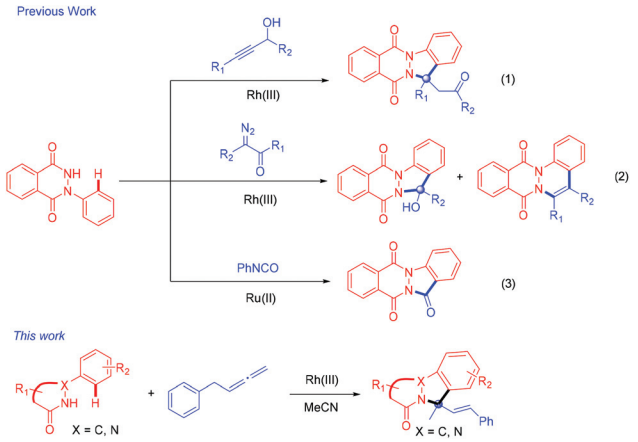


Fig. 1 Some pharmaceutical scaffold motifs.

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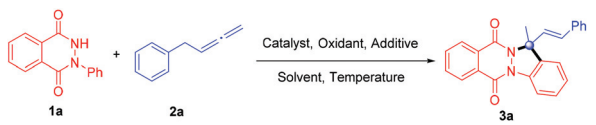
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Scheme 1 Synthesis of indazole derivatives.

Table 1 Optimization of the reaction conditions[a]

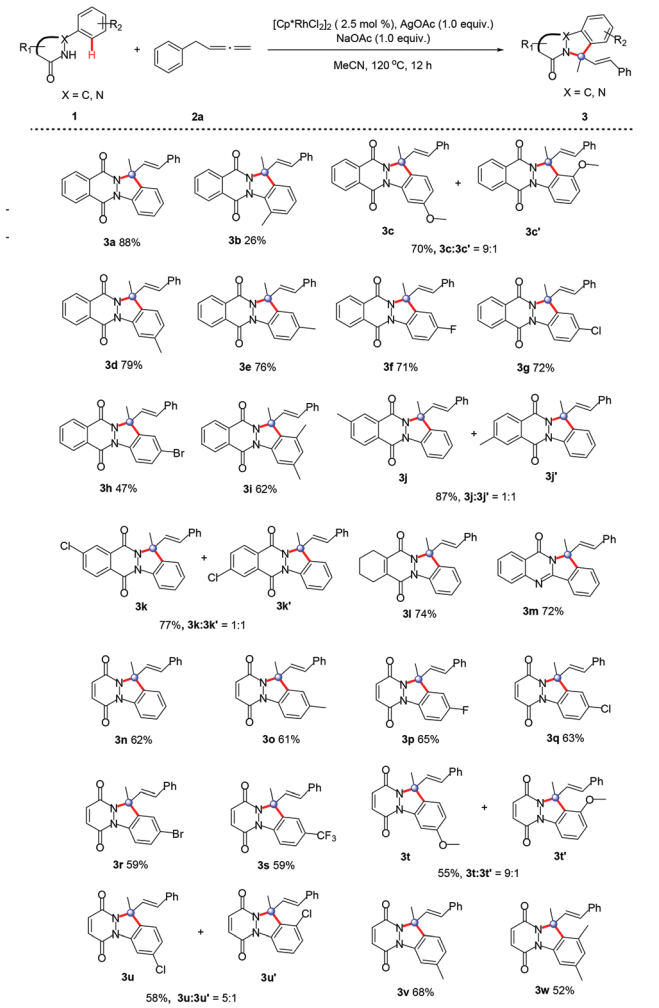


Entry	Catalyst	Oxidant	Additive	Solvent	Yield[b] (%)
1	$[Cp^*RhCl_2]_2$	AgOAc	—	CH_3CN	83
2	$[Cp^*IrCl_2]_2$	AgOAc	—	CH_3CN	35
3	[Ru(*p*-cymene)$Cl_2]_2$	AgOAc	—	CH_3CN	Trace
4	$[Cp^*RhCl_2]_2$	Cu$(OAc)_2$	—	CH_3CN	52
5	$[Cp^*RhCl_2]_2$	Ag_2CO_3	—	CH_3CN	67
6	$[Cp^*RhCl_2]_2$	AgOAc	NaOAc	CH_3CN	85
7	$[Cp^*RhCl_2]_2$	AgOAc	CsOAc	CH_3CN	55
8	$[Cp^*RhCl_2]_2$	AgOAc	K_2HPO_4	CH_3CN	82
9	$[Cp^*RhCl_2]_2$	AgOAc	K_3PO_4	CH_3CN	45
10	$[Cp^*RhCl_2]_2$	AgOAc	Na_2CO_3	CH_3CN	65
11	$[Cp^*RhCl_2]_2$	AgOAc	NaOPiv	CH_3CN	74
12	$[Cp^*RhCl_2]_2$	AgOAc	NaOAc	DCE	46
13	$[Cp^*RhCl_2]_2$	AgOAc	NaOAc	THF	35
14	$[Cp^*RhCl_2]_2$	AgOAc	NaOAc	Toluene	42
15	$[Cp^*RhCl_2]_2$	AgOAc	NaOAc	1,4-Dioxane	27
16	$[Cp^*RhCl_2]_2$	AgOAc	NaOAc	HFIP	24
17[c]	$[Cp^*RhCl_2]_2$	AgOAc	NaOAc	CH_3CN	84
18[d]	$[Cp^*RhCl_2]_2$	AgOAc	NaOAc	CH_3CN	85
19[e]	**$[Cp^*RhCl_2]_2$**	**AgOAc**	**NaOAc**	**CH_3CN**	**88**

[a] Reaction conditions: **1a** (0.20 mmol), **2a** (0.30 mmol), $[Cp^*RhCl_2]_2$ (2.5 mol%) and AgOAc (1.0 equiv.) in MeCN (1.0 mL) were stirred at 100 °C for 12 h under air. [b] Isolated yield. [c] AgOAc (2.0 equiv.). [d] $[Cp^*RhCl_2]_2$ (5.0 mol%). [e] 120 °C.

desired product **3a** was obtained in 83% yield (entry 1). To improve the yield, screening various transition metal catalysts revealed that $[Cp^*RhCl_2]_2$ was slightly favored as a catalyst for the reaction (entries 2 and 3). However, lower yields were obtained when $Cu(OAc)_2$ and Ag_2CO_3 were used as oxidants (entries 4 and 5). To further explore the feasibility of our plot, various additives were then tested (entries 6–11), showing that NaOAc was more appropriate, providing the desired product **3a** in 85% yield (entry 6). In addition, with the use of DCE, THF, toluene, 1,4-dioxane, and HFIP as the solvent, this reaction gave **3a** in lower yields (entries 12–16). Additionally, we failed to obtain a higher yield by changing the amount of the catalyst or oxidant (entries 17 and 18). To our delight, upon increasing the reaction temperature to 120 °C, the yield of **3a** was further enhanced to 88% (entry 19). Thus, the optimized reaction conditions were as follows: **1a** (0.2 mmol), **2a** (0.3 mmol), $[Cp^*RhCl_2]_2$ (2.5 mol%), AgOAc (1.0 equiv.) and NaOAc (1.0 equiv.) in MeCN (1.0 mL) at 120 °C for 12 hours under an air atmosphere.

Based on the above establishment of the optimal reaction conditions, we then investigated the substrate scope of *N*-aryl phthalazinones and pyridazinones. The substrates with various electron-donating or electron-withdrawing groups at different positions could smoothly undergo [4 + 1] cyclization and the fused cyclic products were isolated in acceptable to excellent yields as shown in Scheme 2. The *ortho*-methyl substituted substrate **1b** was affected by steric hindrance, and the



Scheme 2 Substrate scope. Reaction conditions: **1** (0.2 mmol), **2a** (0.3 mmol), $[Cp^*RhCl_2]_2$ (2.5 mol%), AgOAc (1.0 equiv.), NaOAc (1.0 equiv.), MeCN (1.0 mL), 120 °C, 12 h, air.

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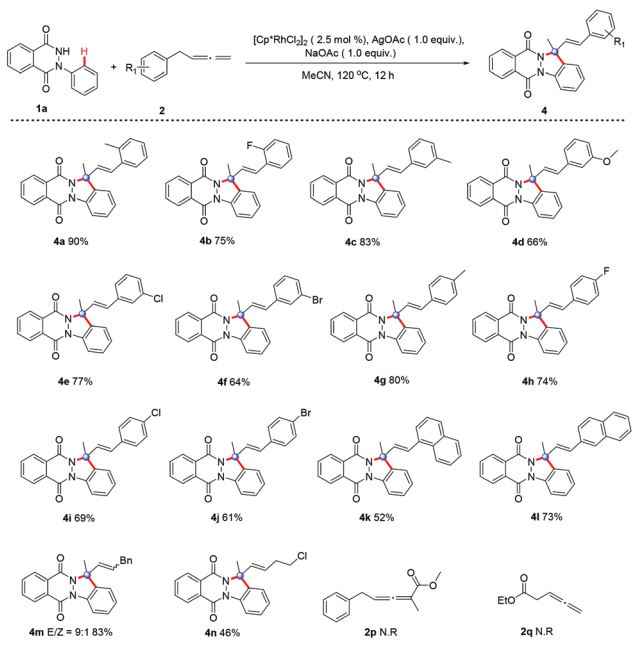
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reaction produced the target product **3b** in a lower yield. Methoxyl substituted **1c** was readily converted to the corresponding regioisomers **3c** and **3c′** in 70% (9 : 1) yield under the optimized conditions. With regard to *m*-Me substituted **1d**, only one isomer **3d** was afforded in 79% yield. When *N*-aryl phthalazinones **1** possess electron-donating or electron-withdrawing groups at the *para* position, such as alkyl (–Me), fluoro (–F), chloro (–Cl) and bromo (–Br) groups, the reaction offered the corresponding products (**3e–3h**) in 47–76% yields under appropriate conditions. The disubstituted substrate **1i** was also well tolerated, forming the desired compound **3i** in 62% yield. When electron-donating groups (–Me) and electron-withdrawing groups (–Cl) were introduced to the phthalazine moiety, smooth conversions were observed to obtain the desired products **3j** and **3k** in 87% and 77% yields, respectively. However, the products **3j** and **3k** were obtained as mixtures of two regioisomers in the same ratio; perhaps their starting substrates **1j** and **1k** were prepared as an inseparable regioisomeric mixture. We were pleased to observe that **3l** and **3m** were obtained in moderate yields. It is worth noting that synthetic derivatives containing an *N*-aryl pyridazinone core are an attractive area due to their broad range of pharmaceutical and agrochemical properties.[15] The substrates with various electron-donating and electron-withdrawing groups at the aryl moiety undergo [4 + 1] cyclization, affording the corresponding products **3n–3w** in 52%–68% yields.

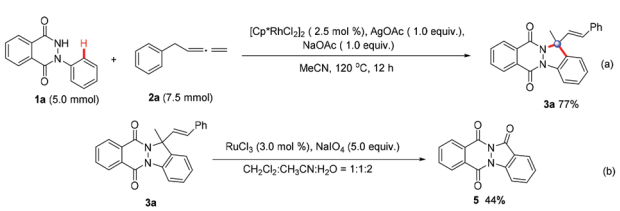
To further expand the versatility of this Rh(III)-catalyzed [4 + 1] annulation reaction, we evaluated the scope of allenes (Scheme 3). It was found that electron-donating (–Me) or electron-deficient (–F) substituents at the 2-position of allene gave the desired products **4a** and **4b** in 90% and 75% yields, respectively. Substrates **2** possessing methyl (–Me), methoxy (–OMe), chloro (–Cl) and bromo (–Br) at the *meta*-position reacted favourably to provide the corresponding products **4c–4f** in 64–83% yields. When methyl (–Me), fluoro (–F), chloro (–Cl) and bromo (–Br) were introduced at the *para*-position, the desired products **4g–4j** were obtained in 61%–80% yields. Similarly, 1-naphthyl- or 2-naphthyl-group-substituted allenes were well tolerated, affording the cyclization products **4k** and **4l** in 52% and 73% yields, respectively. When a long chain aryl substituted substrate **2n** was used, the target product **4m** was obtained with good selectivity (the *E*/*Z* ratio was determined by ^{1}H NMR, *E*/*Z* = 9 : 1) in 83% yield. Pleasantly, aliphatic allene was suitable for the catalytic system and formed the desired compound **4n** in 46% yield, indicating that conjugated olefin as a substrate is not a required driver for the catalytic system. However, the present reaction failed to afford the desired annulation products with 1,1,4-trisubstituted and simple aliphatic allenes containing an ester group.

The synthetic effectiveness of the reaction was proved by performing a gram-scale reaction between **1a** and **2a** to obtain **3a** in 77% yield (Scheme 4a). Additionally, we also tried to conduct a derivatization study of product **3a** to further explore the synthetic value of the reaction. Product **3a** could react with $NaIO_4$ in the presence of $RuCl_3$, obtaining the corresponding product **5** (Scheme 4b).

In order to gain deeper insight into the reaction mechanism, various control experiments were performed (Scheme 5). First, a remarkable deuterium incorporation (55% D) in the *ortho*-C–H bonds of **1e-*d*$_2$** was observed in the presence of CD_3OD. The reaction of **1a** with **2a** in the presence of CD_3OD for 12 h led to the formation of **deuterio-3a** with the deuterium incorporation at the *ortho*-position of the *N*-aryl ring (10% D) and the methyl group of product **3a** (20% D). These results indicated that the C–H bond cleavage step might be reversible and the Rh–H intermediate was formed in the reaction (Scheme 5a). In addition, the kinetic isotope effect values were found to be 2.3 from competition experiments and two parallel runs (Scheme 5b). These values suggest that the C–H bond cleavage is probably involved in the turnover-limiting step. Afterward, an equimolar amount of **1e** and **1f** was used, affording **3e** and **3f** with a ratio of 1.3 : 1 (Scheme 5c). The result showed that an electron-donating group tended to favor the reaction. Moreover, radical inhibition studies have shown that the experiment does not involve radical processes (Scheme 5d).

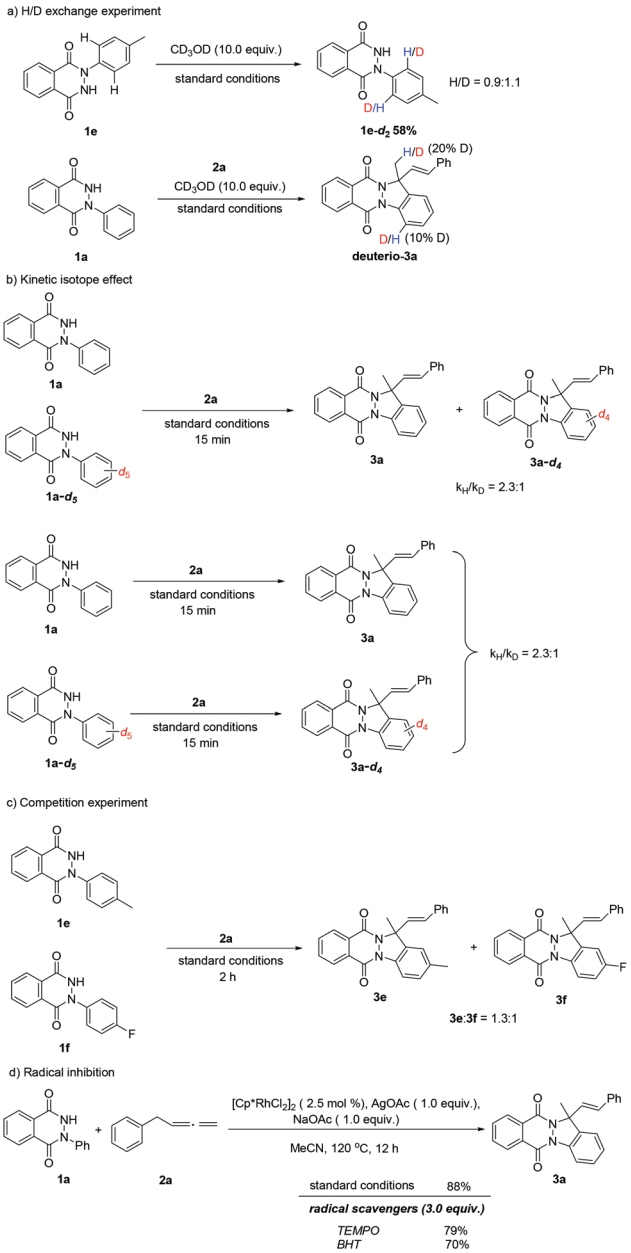


Scheme 3 Substrate scope of allenes. Reaction conditions: **1a** (0.2 mmol), **2** (0.3 mmol), $[Cp^*RhCl_2]_2$ (2.5 mol%), AgOAc (1.0 equiv.), NaOAc (1.0 equiv.), MeCN (1.0 mL), 120 °C, 12 h, air.



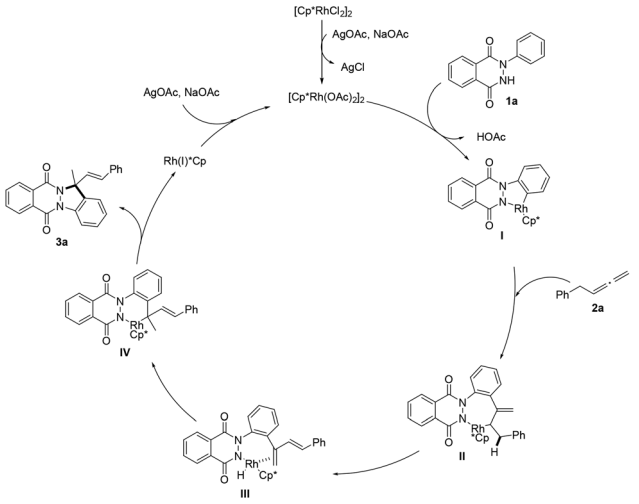
Scheme 4 Gram scale synthesis and derivatization.

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Scheme 5 Mechanistic studies.

Based on our preliminary mechanistic studies and the previous literature,[8c,12,16] a plausible mechanism for Rh(III)-catalyzed [4 + 1] annulation of *N*-aryl phthalazinones and pyridazinones with allenes is described in Scheme 6. First, substrate **1a** reacts with active rhodium, produced *in situ* through ion exchange of $[Cp^*RhCl_2]_2$ with AgOAc and NaOAc, to form intermediate **I**. Then, insertion of allene **2a** followed by β-hydride elimination offers intermediate **III**. The Rh–H in intermediate **III** is inserted into the diene moiety to constitute **IV**, which then undergoes reductive elimination to give compound **3a**. The Rh(I) species generated from the catalytic cycle is reoxidized by AgOAc and NaOAc to give active rhodium.



Scheme 6 Plausible mechanism.

Conclusions

In conclusion, we have successfully developed a Rh(III)-catalyzed C–H oxidative cyclization of *N*-aryl phthalazinones and pyridazinones with a variety of allenes. A variety of indazole derivatives were constructively synthesized in moderate to excellent yields. This catalytic system exhibits excellent functional group tolerance and atom economy. Further evaluation of the suitability of this catalytic system is currently being carried out in our laboratory.

Experimental section

The general procedure for the C–H/N–H activation of *N*-aryl phthalazinone **1a** with allene **2a** is as follows. A mixture of 2-phenyl-2,3-dihydrophthalazine-1,4-dione **1a** (47.6 mg, 0.20 mmol), buta-2,3-dien-1-ylbenzene **2a** (39.1 mg, 0.30 mmol), $[Rh^*CpCl_2]_2$ (3.1 mg, 2.5 mol%), NaOAc (16.4 mg, 0.20 mmol) and AgOAc (33.4 mg, 0.20 mmol) in MeCN (1.0 mL) was stirred at 120 °C under open air for 12 hours. The resulting mixture was cooled to room temperature, the solvent was removed under vacuum and the crude reaction mixture was purified by silica gel (200–300 mesh) column chromatography using hexane/ethyl acetate (6 : 1) as the eluant to afford **3a** as a yellow solid.

Author contributions

Chuanming Yu conceived this work. Chuanliu Yin, Tianshuo Zhong and Xiangyun Zheng performed the experiments. Lihao Li and Jian Zhou analyzed the data and prepared the ESI. Chuanliu Yin and Xiangyun Zheng wrote this manuscript. All authors read and approved the final version to be published.

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Conflicts of interest

There are no conflicts to declare.

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